Memory Decay in LLM Agents: Cognitive Foundations, Graph Implementations, and System Architecture

Executive Summary The integration of cognitively plausible memory decay into multi-agent simulations requires a paradigm shift from static temporal knowledge graphs to dynamic, retention-scored architectures. The recommended decay function is the power-law-based Retrievability equation from the Free Spaced Repetition Scheduler (FSRS), defined as $R(t, S) = (1 + t/(9S))^{-1}$, which natively incorporates the spacing effect and avoids the statistical artifacts of exponential decay models. The minimal graph schema extends Graphiti's bi-temporal properties (valid_at, invalid_at) with cognitive metadata: stability, difficulty, last_accessed, access_count, and valence. To prevent database write contention, retrieval scoring must employ lazy evaluation, calculating current retrievability dynamically at query time by combining semantic similarity, importance, and R. Hard deletion should be avoided in favor of archiving; scheduled jobs should be strictly reserved for semantic consolidation and the enforcement of absolute storage budgets. This recommendation would change only if graph traversal latency under lazy evaluation strictly violates real-time simulation bounds, necessitating a fallback to quantized, batch-computed decay tiers.

1. Shape of Forgetting

a. Exponential vs. Power-Law vs. Other Forms

The exact mathematical formulation of human forgetting has been intensely debated in cognitive psychology since Hermann Ebbinghaus introduced the exponential forgetting curve in 1885. The traditional Ebbinghaus model posits that memory retention drops logarithmically, modeled fundamentally as an exponential decay function, $R = e^{-t/S}$. However, modern cognitive modeling strongly favors the power law of forgetting, typically expressed as $R = at^{-b}$.

(Strongly supported): The literature reports that individual human memory retention functions are significantly better described by a power function than by an exponential function[1]. Wixted and Ebbesen (1997) definitively demonstrated that while the arithmetic averaging of multiple memory items with varying decay rates can mathematically distort aggregate data to appear exponential, the fundamental, item-level decay follows a power law[1]. Their quantitative analysis of individual subject forgetting functions proved that geometric averaging avoids this artifact and preserves the underlying power-law shape[3].

(Mixed): Some computational models suggest a Weibull distribution or a hybrid exponential-power function. However, Bayesian model selection often favors the power function, and these hybrid forms are frequently mathematically indistinguishable from pure power functions given the typical noise present in human recall data[2].

(Strongly supported): The Adaptive Control of Thought-Rational (ACT-R) cognitive architecture, an integrated theory of the mind, models base-level activation (memory strength) using a power-law formulation: $A_i = \ln \sum_{j=1}^{n} t_j^{-d}$, where t_j is the time since the j -th practice and d is the decay rate (typically around 0.5)[6]. This aligns with the observation that older memories decay more slowly than newer ones, a phenomenon the exponential model fails to capture accurately.

System Recommendation: Agents should utilize a power-law-based decay function. The FSRS (Free Spaced Repetition Scheduler) algorithm provides a computationally efficient power-law retrievability equation: $R(t, S) = (1 + t/(9S))^{-1}$, where t is the elapsed time in days and S is memory stability[8]. This avoids the sharp, unrealistic cliff of exponential decay and models the "long tail" of memory persistence accurately, making it ideal for a social simulation where agents must occasionally recall distant past events without keeping them constantly in active working memory.

b. The Spacing Effect and Repeated Retrieval

The spacing effect dictates that memory retention improves dramatically when encoding events are distributed over time rather than massed together. Every successful retrieval of a memory reconsolidates it, increasing its stability and reducing its future rate of decay.

(Strongly supported): Retrieval is not a passive read operation; it is a learning event that alters the memory trace, a phenomenon known as the testing effect established by Roediger and Karpicke[10].

(Strongly supported): In the FSRS algorithm, memory is modeled via three state variables: Difficulty (D), Stability (S), and Retrievability (R)[8]. FSRS defines stability as the exact time interval (in days) required for retrievability to drop from 100% to 90%[10]. Upon a successful recall, the new stability S' increases based on the current difficulty and stability. FSRS models the stabilization curve by acknowledging that the lower the retrievability at the time of recall, the higher the resulting stability increase, provided the recall is successful[9]. This algorithm achieves an 81% better predictive accuracy over the older SuperMemo-2 (SM-2) algorithm[10]. Duolingo's Half-Life Regression (HLR) models a similar dynamic but uses a more simplistic formulation that is less optimized for individualized sequential review history[11].

For LLM agents, a retrieval event during graph vector search should trigger a "reinforcement" update, artificially treating the retrieval as a spaced repetition review, thereby increasing S and preserving heavily utilized facts against the relentless pressure of time.

2. Emotional Salience

a. Valence and Arousal Effects on Retention

Emotional memories are consolidated more strongly than neutral memories, a process biochemically mediated by the amygdala interacting with the hippocampus during memory encoding. The memory modulation literature strictly divides emotion into two orthogonal

dimensions: valence (the spectrum from positive to negative) and arousal (the spectrum from calm to excited).
(Strongly supported): High-arousal memories are consolidated more efficiently and decay significantly slower, regardless of valence. This is known as the arousal-dependent enhancement effect.
(Mixed): The literature is divided on whether positive or negative memories fade faster. The Fading Affect Bias (FAB) describes the robust psychological tendency for the negative affect associated with unpleasant memories to fade faster than the positive affect of pleasant memories, allowing humans to maintain psychological resilience. Furthermore, the "positivity effect" demonstrates that older adults tend to preferentially retain positive memories. Conversely, evolutionary "negativity bias" ensures that traumatic or dangerous events (high arousal, extreme negative valence) are seared into memory to prevent future harm.

b. Flashbulb Memories

(Strongly supported): Highly significant, surprising, and emotionally arousing events trigger "flashbulb memories," a term coined by Brown and Kulik. While human subjects report extremely high confidence in the vividness and accuracy of these memories over decades, objective longitudinal testing reveals that their factual accuracy decays at a rate similar to ordinary episodic memories. Thus, high emotional salience artificially inflates retrieval confidence and subjective vividness without perfectly preserving semantic accuracy.

c. Weighting Scheme for Agent Architectures

(Inference): Because agent systems evaluate memory based on computational utility and graph weights rather than biological subjective confidence, emotional salience should directly modulate the initial stability (S_0) and initial difficulty (D_0) of a memory node at encoding time.

Proposed Weighting Formula: Let arousal $A \in [0, 1]$ and valence $V \in [-1, 1]$. Let the baseline FSRS parameters for a neutral memory be $S_{base} = 2.0$ and $D_{base} = 5.0$.

- Initial Stability: $S_0 = S_{base} \times (1 + \alpha A) \times (1 + \beta |V|)$
- Initial Difficulty: $D_0 = \max(1.0, \min(10.0, D_{base} - \gamma A - \delta V_{penalty}))$

Where $\alpha \approx 1.5$ (arousal weight), $\beta \approx 0.5$ (valence magnitude weight), and $V_{penalty}$ is applied strongly for negative valence to represent the cognitive difficulty of processing trauma. To simulate the Fading Affect Bias, the system can apply a linear temporal decay specifically to the stored valence attribute of negative memories, moving it toward neutral, independent of the factual retrievability R of the node itself.

3. Consolidation and Reconsolidation

a. Sleep, Rehearsal, and Reconsolidation Mechanisms

(Strongly supported): Human memory relies on complementary learning systems. The hippocampus rapidly encodes initial episodic memory traces during daily experiences. These temporary traces are then slowly transferred to the neocortex during offline periods (such as

sleep or quiet rest) to form generalized, stable semantic representations.*(Strongly supported)*: Nader's work on reconsolidation proved that when a memory is recalled, it does not simply read out statically; it enters a labile state where it must be reconsolidated into the network. This lability provides a critical window where the memory can be updated, distorted by new context, or weakened.

b. Episodic to Semantic Conversion

Over time, the brain extracts the "gist" of repeated episodic experiences to form semantic schemas. The individual episodic details (the specific time and place) decay rapidly, but the extracted semantic rules persist indefinitely. Bahrick's extensive longitudinal research on "permastore" indicates that certain semantic memories, after an initial decay phase of 3 to 6 years, reach a remarkably stable plateau where they do not decay further for up to 50 years[12].

c. Implementation in Agent Systems

(Strongly supported): Existing advanced agent architectures implement consolidation via periodic offline reflection and tier migration. Generative Agents utilize a reflection mechanism that routinely summarizes lower-level episodic memories into higher-level semantic insights, forming a tree-like memory structure.

(Strongly supported): MemoryOS employs a rigorous three-tier architecture inspired by operating systems: Short-Term Interaction Memory (STIM) utilizing LRU eviction, Mid-Term Episodic Memory (MTEM), and Long-Term Semantic Memory (LTSM) for compressed knowledge[16]. A-Mem (Agentic Memory) utilizes Generative Reward Policy Optimization (GRPO) to teach a specialized Consolidator agent how to compress fragmented historical facts into robust semantic representations during simulated "dream cycles"[18]. Mem0 similarly executes an update stage where an LLM extracts entities and relationships from conversational turns, integrating them into the broader knowledge graph[19].

(Recommendation): Agents operating on a Graphiti/Neo4j backend should implement consolidation through an asynchronous background graph-summarization job (analogous to the "Dreamer" cycle in A-Mem)[16]. Episodic event nodes (e.g., "Agent A insulted Agent B at the tavern today") should be periodically queried. An LLM extracts the semantic gist (e.g., "Agent A holds a grudge against Agent B"), creates a new Semantic edge connecting the entity nodes, and applies a heavy decay penalty to the original episodic nodes, allowing them to fade gracefully while the semantic knowledge enters a high-stability "permastore" state.

4. Interference and Retrieval Failure

a. Proactive and Retroactive Interference

(Strongly supported): Robert Bjork's New Theory of Disuse fundamentally reshaped the understanding of forgetting by proposing that a memory is never truly lost from storage; rather, it becomes inaccessible due to a loss of retrieval strength. Forgetting is primarily driven by interference: proactive interference (old, entrenched memories disrupting the retrieval of new, similar ones) and retroactive interference (new memories overwriting access to old, similar ones). Furthermore, memory retrieval is highly cue-dependent; the loss of the original contextual retrieval cues makes the memory inaccessible.

(Mixed): The debate over whether passive temporal decay exists entirely independent of

interference remains actively contested in human experimental psychology. However, in computational architectures, an explicit temporal decay parameter is almost universally required to maintain system viability and computationally bound the retrieval search space.

b. Design Implications for LLM Agents

(Inference): Because a memory is theoretically inaccessible rather than erased, agent architectures should strictly avoid hard-deleting low-weight nodes during routine operations. Instead, they should archive or down-rank them based on retrievability scores. If a highly specific new context arises (providing a strong semantic retrieval cue), the mathematically decayed memory can still be recovered via vector similarity bridging the gap. Hard-deletion (purging) should be reserved strictly for global storage budget enforcement, GDPR-style privacy compliance, or adversarial memory-attack mitigation[20].

5. Memory Types and Their Decay

The following table categorizes memory types for social-simulation agents, synthesizing cognitive science literature with LLM architecture requirements to dictate decay parameters.

Memory Type	Description	Relative Decay Rate	What Slows Decay	What Accelerates Decay	Evidence Strength
Episodic	Discrete events bound to a specific time and place (e.g., a conversation).	Very Fast	High emotional arousal, subsequent repeated retrieval/rehearsal.	Routine repetition of highly similar events (retroactive interference).	Strongly Supported
Semantic	Facts, rules, and general impressions extracted from episodes.	Slow	Consolidation cycles, entry into permastore, high graph connectivity.	Contradictory new facts, paradigm shifts in the simulation.	Strongly Supported[13]
Procedural	Habits, styles, learned	Extremely Slow	Repeated execution, continuous	Extended periods of absolute	Strongly Supported

	behaviors (e.g., specific writing styles).		reinforcement learning loops.	disuse.	
Emotional	Affective associations bound to entities (e.g., fear of a rival agent).	Slow	High initial arousal, extreme negative valence (survival bias).	Extinction training (repeated exposure without negative consequence).	Mixed
Social	Relationships, norms, interaction histories, and network graphs.	Moderate	Frequent direct interaction, social validation from peers.	Isolation, relationship invalidation, betrayal.	Strongly Supported[20]
Goals	Current commitments and immediate task objectives.	Variable (Task Bound)	High priority, dependencies, frequent plan checking.	Task completion, supersession by higher priority goals.	Strongly Supported[20]

6. Existing LLM-Agent Forgetting Policies

a. System Comparison

System	Forgetting Mechanism	Decay / Threshold Parameters	Retention Triggers	Evaluation & Results
MaRS (Memory-Aware Retention	6 configurable policies (FIFO, LRU, Priority	Bound by a global budget	Temporal access indices, preservation of	Evaluated on the FiFA Benchmark.

Schema)	Decay, Reflection, Random, Hybrid).	B. Utility optimized via a submodular knapsack formulation[20].	node dependencies (provenance).	The Hybrid policy scored $\approx$ composite[20].
Mem0 (Graph Variant)	Hard pruning via a composite scoring threshold[19].	Retains nodes if score $>$. Score combines recency, frequency, degree centrality, and age[19].	Checked every 400 turns. Deletes nodes and incident edges[19].	LoCoMo Benchmark. Pruned 9.8% of nodes with negligible F1 impact, bounding graph growth[22].
MemoryOS	Tiered hierarchical eviction and heat-based routing[16].	LRU eviction from STIM; continuous access/recency heat scoring for LTSM[23].	Frequent retrievals increase the access-and-recency heat[16].	Demonstrated average improvement across tasks on the LoCoMo benchmark[16].
A-Mem (Agentic Memory)	Soft decay via prediction and reversible demotion[16].	Node demotion factor γ. $\gamma =$ is hard deletion. Decay lowers retrieval rank[16].	Later dream cycles revealing renewed query relevance restore the node[16].	MemoryAgent Bench (MAB). Soft decay improved long-horizon answer F1 by avoiding permanent loss[25].

b. Efficacy of Purposeful Forgetting vs. Verbatim Retention

(Strongly supported): Empirical benchmarks definitively demonstrate that keeping all facts indefinitely degrades performance. On the FiFA benchmark, absent explicit retention control, computational costs rise while narrative coherence and social recall degrade as the context window becomes cluttered with proactive and retroactive interference[20]. Similarly, the Mem0 graph-variant evaluation showed that applying a selective forgetting module that removed 9.8% of nodes did not materially hurt token F1 but successfully prevented memory bloat[22].
(Contested): The specific mechanism of removal remains debated. Mem0's paper uses active

deletion (purge) for low-scoring nodes[19], whereas A-Mem champions soft decay (demotion) that is reversible if future dream cycles discover new relevance[16]. Adversarial testing on the ForgetEval framework highlights that systems focusing solely on the "recall plane" completely fail control-plane tests where agents must strictly forget superseded or deprecated data (e.g., rotated passwords)[21].

c. Known Risks of Forgetting Policies

The primary risks of purposeful forgetting include:

1. **Loss of Dependency (Orphaned Facts)**: Forgetting a foundational episodic premise while retaining the generalized semantic conclusion can lead to hallucinations when the agent is challenged to justify its beliefs. MaRS mitigates this by enforcing provenance structures, utilizing depends_on edges to strictly block the deletion of prerequisites[20].
2. **Narrative Incoherence**: Agents may blatantly contradict their past behaviors or stated preferences if procedural or social memories are aggressively pruned to save context space.
3. **Privacy Leakage**: Inadequate forgetting of sensitive cross-session data can lead to data contamination between users, a major issue highlighted in GDPR right-to-be-forgotten contexts[21].

7. Mapping to a Graph Database (Graphiti / Neo4j)

a. Proposed Schema and Temporal Semantics

Graphiti models time fundamentally using bi-temporal semantics: valid_at and invalid_at[26]. It is crucial to distinguish this from cognitive decay. invalid_at denotes when a fact ceases to be true in the simulated world (e.g., Agent A is no longer employed by the Tavern). This is an absolute logical state, not a representation of human forgetting. A fact can be entirely logically valid but cognitively forgotten by the agent, or logically invalid but vividly remembered (e.g., remembering a past job).

Required Node & Edge Properties: To implement FSRS-based decay natively in Graphiti, the schema requires the following minimal property set on all memory nodes:

- created_at: System timestamp of physical graph insertion.
- valid_at: World-time the fact became logically true.
- invalid_at: World-time the fact became logically false (null if still true).
- last_accessed: System timestamp of the most recent retrieval.
- access_count: Integer counting total historical retrievals.
- stability (S): FSRS memory stability parameter (float, default 2.0).
- difficulty (D): FSRS memory difficulty parameter (float, default 5.0).
- valence: Emotional alignment $[-1.0, 1.0]$.
- arousal: Emotional intensity $[0.0, 1.0]$.
- memory_type: Categorical tag (Episodic, Semantic, Social, etc.).
- consolidation_state: Enum (Raw, Summarized, Archived).

b. Scheduled vs. Lazy Decay

(Inference): Rewriting all node weights via a scheduled chronological cron job (Scheduled Decay) forces continuous write-locks across the database. In Neo4j, updating properties creates new node versions, causing massive write contention, MVCC (Multi-Version Concurrency Control) bloat, and severe scaling bottlenecks.
Instead, **Lazy Decay** computes the effective memory strength dynamically at retrieval time. The database only stores the metadata (last_accessed, S, D).

Logic for Lazy Decay at Retrieval (Pseudocode for Cypher): At query time T_{now}, calculate elapsed time $t = T_{\text{now}} - \text{last_accessed}$. Calculate instantaneous Retrievability $R = (1 + t/(9S))^{-1}$. Combine R with the semantic vector similarity score to rank the final retrieval results, completely avoiding continuous background writes.

c. Invalidation vs. Fading

(Strongly supported): As stated, edge invalidation alters the logical truth of the graph, while fading (decay) alters the cognitive accessibility of the graph. Retrieval logic must respect this separation. A standard query must first filter for validity (WHERE invalid_at IS NULL) to ensure it only retrieves currently true facts, and *then* sort those true facts by retrievability R. Historical queries ("What did you used to believe?") invert the validity filter but maintain the retrievability sort.

d. Scaling Considerations for Multi-Agent Simulations

For simulations with thousands of agents sharing a single Neo4j instance, memory must be partitioned. While Graphiti supports tenant boundaries[28], a shared world requires isolated cognitive subgraphs per agent, connected by BELONGS_TO edges to a central Agent node, to prevent cross-contamination. Write contention from last_accessed updates during retrieval is a massive scaling risk. To mitigate this, updates should be batched. Instead of writing to Neo4j on every single read, an in-memory cache layer (e.g., Redis/FalkorDB) should aggregate read events, executing bulk Cypher UNWIND updates to Neo4j every 100 simulation rounds. Vector indices must be carefully constructed to pre-filter by the Agent ID to maintain retrieval latency under 100ms.

8. Integration with Existing Mechanisms

The requested custom mechanisms map elegantly onto the FSRS DSR model, creating one unified mathematical framework rather than disparate edge cases:

- **Boredom / Fatigue (Inhibition of Return)**: When audience agents observe repeated styles, instead of a permanent alteration, a short-term, transient penalty is applied to the retrievability R at query time (independent of S). Alternatively, repetitive access without meaningful task utility increases the D (Difficulty) parameter slightly, causing the memory's future stability to grow much slower.
- **Failure Memory (Flops)**: A real-world flop is encoded with high arousal and extreme negative valence. This emotional salience massively increases the initial stability S_0,

making it very hard to forget. When this memory is retrieved during similarity search for a new idea, the negative valence attribute acts as a penalty modifier to the downstream action utility function, ensuring similar ideas are "met coldly."

- **Success Memory (Wins)**: A win is encoded with high positive valence and high arousal. Successful real-world execution acts as a massive spaced-repetition reinforcement event, sharply dropping Difficulty D (it is an "Easy" grade) and multiplying Stability S by a success-factor (e.g., $1.5\times$ to $2.0\times$).
- **Demotion Ratchet**: Falling out of favor is not standard temporal decay; it is an active invalidation of status. This is modeled as a "Lapse" in FSRS. When an idea is demoted, a reinforcement event is fired with a "Fail" grade, triggering the FSRS post-lapse formula: $S_{\text{new}} \approx \min(S_{\text{old}}, \text{decay_factor} \times e^{\text{lapses}})$. This immediately crushes the stability, making it mathematically harder to climb back after each subsequent fall, exactly as requested.

9. Personality Interaction

(Mixed / Inference): The cognitive psychology literature explicitly links Big Five personality traits to memory biases. Neuroticism is robustly correlated with rumination—the repeated, involuntary retrieval of negative episodic memories. In an agent system, an agent's trait profile can programmatically modulate decay parameters:

- **High Neuroticism**: The system automatically generates stochastic "phantom" access events for memories possessing negative valence, artificially keeping their S high and elapsed time t low, mathematically simulating rumination.
- **High Openness**: Lowers the initial D (difficulty) parameter for novel Semantic memories, allowing them to consolidate and stabilize faster.
- **Conscientiousness**: Decreases the baseline decay rate specifically for Goal-type and Task-type memories, ensuring long-term commitments do not fade.

(Note: While the psychological phenomena are strongly supported, adapting them as math modifiers on FSRS variables for LLM agents is a novel architectural inference.)

10. Validation

To empirically validate that the agent's forgetting is human-like and computationally safe, the system must employ a multi-tiered testing strategy:

1. **Behavioral Retention Curves**: Measure the agent's ability to recall a specific injected fact over simulated time. Plotting the probability of recall success must reveal a power-law distribution, avoiding the exponential cliff, and must demonstrate savings (faster relearning of forgotten facts).
2. **FiFA Benchmark Dimensions**: Evaluate the agents on Narrative Coherence (does the agent contradict past actions?), Goal Completion (does it forget active tasks?), and Social Recall Accuracy (does it recognize established relationships?)[20].
3. **Ablation Studies**: Compare the FSRS-decay agent against a baseline of "No Forgetting"

(infinite context window) and "Hard-Cutoff" (a sliding window LRU cache).*Acceptance Thresholds*: The FSRS decay model must maintain $> 90\%$ Narrative Coherence compared to the No-Forgetting baseline while systematically reducing token context size and inference costs by at least 40%.

11. Privacy and Safety

The MaRS architecture introduces Privacy-Decay, which links memory retention directly to data sensitivity[20]. In a social simulation where agents might share a unified graph backend, differential privacy can be injected by adding Laplacian noise to the retrievability score of highly sensitive memories (e.g., secrets). This causes sensitive data to become stochastically inaccessible. Furthermore, if GDPR-style "Right to Be Forgotten" compliance is simulated, soft decay is insufficient. A targeted hard purge operation must physically delete the sensitive node and recursively delete all downstream Semantic nodes that possess a depends_on provenance edge pointing to the purged node[20].

12. Recommended Architectures

Recommended Minimal Design (Lazy Decay)

The minimal design avoids Python-side scheduled jobs entirely. It calculates decay at retrieval time using Cypher, keeping the architecture lean and cost-effective.

Cypher Implementation (Retrieval with Lazy Decay):

```
Cypher
// Retrieve valid facts, calculate FSRS retrievability, and combine with vector similarity
MATCH (a:Agent {id: $agent_id})-[:REMEMBERS]->(m:Memory)
WHERE m.invalid_at IS NULL // Logical validity filter
WITH m,
     duration.inDays(m.last_accessed, datetime()).days AS t,
     m.stability AS S
// FSRS Retrievability Formula
WITH m, (1.0 + (t / (9.0 * S))) ^ -1.0 AS retrievability
WHERE retrievability > 0.15 // Soft threshold to prune the long tail
// Compute final score blending similarity, retrievability, and emotional importance
WITH m, retrievability,
     gds.similarity.cosine(m.embedding, $query_embedding) AS semantic_sim
WITH m, (0.6 * semantic_sim) + (0.2 * retrievability) + (0.2 * abs(m.valence)) AS final_score
ORDER BY final_score DESC
LIMIT 10
RETURN m.text, m.memory_type, final_score
```

Full Design (FSRS Integrated Graph with Batching)

The full design incorporates lazy retrieval, asynchronous batch reinforcement, and semantic consolidation jobs.

Python Implementation (Neo4j Backend):

```
Python
import math
from datetime import datetime

class CognitiveMemoryManager:
    def __init__(self, driver):
        self.driver = driver

    def calculate_initial_dsr(self, valence, arousal):
        """Calculates initial Difficulty and Stability based on emotional salience."""
        # Baseline FSRS values
        D0 = 5.0
        S0 = 2.0

        # High arousal boosts stability, extreme valence alters difficulty
        S_new = S0 * (1.0 + 1.5 * arousal) * (1.0 + 0.5 * abs(valence))
        D_new = max(1.0, min(10.0, D0 - (abs(valence) * 2.0)))
        return D_new, S_new

    def add_memory(self, agent_id, text, valence, arousal, memory_type):
        d, s = self.calculate_initial_dsr(valence, arousal)
        query = """
        MATCH (a:Agent {id: $agent_id})
        CREATE (m:Memory {
            text: $text,
            valid_at: datetime(),
            created_at: datetime(),
            last_accessed: datetime(),
            access_count: 1,
            difficulty: $d,
            stability: $s,
            valence: $valence,
            arousal: $arousal,
            type: $type
        })
        CREATE (a)-[:REMEMBERS]->(m)
```

```
        """
        with self.driver.session() as session:
            session.run(query, agent_id=agent_id, text=text, d=d, s=s,
                        valence=valence, arousal=arousal, type=memory_type)

    def batch_reinforce_memories(self, memory_ids, grade="Good"):
        """Asynchronously updates DSR parameters for memories retrieved during the round."""
        # Simplified FSRS stability increase approximation for successful retrieval
        query = """
        UNWIND $memory_ids AS mem_id
        MATCH (m:Memory) WHERE id(m) = mem_id
        // Record current retrievability before update
        WITH m, duration.inDays(m.last_accessed, datetime()).days AS t
        WITH m, (1.0 + (t / (9.0 * m.stability))) ^ -1.0 AS R

        SET m.access_count = m.access_count + 1,
            m.last_accessed = datetime(),
            // FSRS stability multiplier based on difficulty and previous R
            m.stability = m.stability * (1.0 + math.exp(0.1 * m.difficulty) * (1.0 - R))
        """
        with self.driver.session() as session:
            session.run(query, memory_ids=memory_ids)

    def prune_and_consolidate(self):
        """Scheduled background job running every N rounds to clean the graph."""
        query = """
        MATCH (m:Memory {type: 'Episodic'})
        WITH m, duration.inDays(m.last_accessed, datetime()).days AS t
        WITH m, (1.0 + (t / (9.0 * m.stability))) ^ -1.0 AS R
        WHERE R < 0.05
        // Soft delete: detach from active memory, attach to Archive
        REMOVE m:Memory
        SET m:ArchivedMemory
        RETURN m.text
        """
        # Followed by an LLM summarization prompt over archived nodes to generate new Semantic nodes
        pass
```

13. Worked Example: 100 Simulation Rounds

Agent: "Kael" (Personality Profile: High Neuroticism) **Round 1:** Kael attempts to perform a magic trick at the tavern. It is a spectacular failure (Failure Memory / Flop).

- **Encoding:** Node 101 created. type: Episodic, valence: -0.9 (humiliation), arousal: 0.8 (embarrassment).

- **Initial DSR:** Due to high arousal and extreme valence, S_0 is calculated at 4.6 (highly stable). Because of High Neuroticism, Kael's background loop stochastically touches this node without a prompt, artificially maintaining a low t.

Round 10: Kael is asked by another agent to perform magic again.

- **Retrieval:** The conversational context "perform magic" triggers a semantic search.
- **Retrievability:** Assuming $t = 9$ days. $R = (1 + 9/(9 \times 4.6))^{-1} \approx 0.82$. The memory is highly accessible.
- **Execution:** The retrieval succeeds. The heavily negative valence applies a massive penalty to the action utility function. Kael angrily refuses to perform.
- **Reinforcement:** The retrieval constitutes a rehearsal event. S is multiplied according to the FSRS formula, increasing to 6.2. last_accessed is updated to Round 10.

Round 50: Background Consolidation Job.

- Node 101 has not been accessed since Round 10 ($t = 40$). $R = (1 + 40/(9 \times 6.2))^{-1} \approx 0.58$.
- The system scans episodic memories. Node 101 is summarized into Node 205: "Kael believes he is terrible at magic." type: Semantic, $S_0 = 10.0$ (Semantic facts have inherently higher base stability). Node 101 is transitioned to ArchivedMemory.

Round 100: Kael is mocked about magic by a rival.

- **Retrieval:** Node 205 (Semantic) has an inherently lower Difficulty D, leading to a massive S value (functioning as Bahrick's Permastore). Despite $t = 50$ since consolidation, $R \approx 0.95$. Kael instantly recalls his belief and reacts defensively. The underlying episodic memory (Node 101) has decayed to $R < 0.1$ and is not retrieved; Kael remembers *that* he hates magic, but can no longer easily picture the specific tavern incident.

14. Open Problems & Unverified Claims

1. **DSR Tuning for LLM Tokens**: The constants in FSRS (e.g., the 9 scalar, $e^{0.1D}$) were empirically optimized on millions of human spaced repetition data points (flashcard reviews)[30]. It remains unverified whether these exact mathematical weights perfectly optimize LLM context window retention. They provide a cognitively plausible behavioral shape, but the parameters may require re-tuning specifically for token-efficiency optimization.
2. **Trait-Modulated Decay Benchmarking**: The programmatic translation of Big Five psychological traits to mathematical decay parameters (e.g., Neuroticism artificially inflating access counts) is an architectural inference based on cognitive psychology literature regarding rumination[31]; its efficacy has not been formally benchmarked in

existing systems like Generative Agents or Mem0.

3. **Graph Vector Query Latency**: While Graphiti documentation explicitly outlines valid_at and invalid_at[27], it lacks native continuous cognitive decay properties. The integration of continuous math into Neo4j via Cypher (e.g., WHERE (1.0 + (t / (9.0 * S))) ^ -1.0 > 0.15) is syntactically valid but will likely bypass vector HNSW index acceleration if poorly constructed. This may necessitate a hybrid retrieval strategy where a smaller top-K vector search is performed first, and post-filtered by retrievability in memory.

References

Alqithami, S. (2025). Forgetful but Faithful: A Cognitive Memory Architecture and Benchmark for Privacy-Aware Generative Agents. *arXiv preprint arXiv:2512.12856*. https://arxiv.org/abs/2512.12856

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of the mind. *Psychological Review*, 111(4), 1036-1060.

Bahrick, H. P. (1984). Semantic memory content in permastore: Fifty years of memory for Spanish learned in school. *Journal of Experimental Psychology: General*, 113(1), 1-29.

Chhikara, P., Khant, D., Aryan, S., Singh, T., & Yadav, D. (2025). Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. *arXiv preprint arXiv:2504.19413*. https://arxiv.org/abs/2504.19413

Du, P., et al. (2026). Memory for Autonomous LLM Agents: Mechanisms, Evaluation, and Emerging Frontiers. *arXiv preprint arXiv:2603.07670*. https://arxiv.org/abs/2603.07670

Hu, Y., Liu, S., Yue, Y., & Zhang, G. (2026). Memory in the Age of AI Agents. *arXiv preprint arXiv:2512.13564*. https://arxiv.org/abs/2512.13564

Kang, J., et al. (2025). MemoryOS. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Repository: https://github.com/BAI-LAB/MemoryOS

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 1-22.

Rusu, T., Khanzadeh, S., & Alalfi, M. (2026). Selective Forgetting: A Graph-Based Memory Framework for Long-Term LLM Agents. *arXiv preprint arXiv:2608.28978*.

Wixted, J. T., & Ebbesen, E. B. (1997). Genuine power curves in forgetting: A quantitative analysis of individual subject forgetting functions. *Memory & Cognition*, 25(5), 731-739.

Xu, W., et al. (2025). A-MEM: Agentic Memory for LLM Agents. *arXiv preprint arXiv:2502.12110*. https://arxiv.org/abs/2502.12110

Yang, D. (2026). Control-Plane Placement Shapes Forgetting: An Architectural Study of Agent Memory Across Thirteen System Configurations. *arXiv preprint arXiv:2606.15903*.

Ye, J. (Open Spaced Repetition / FSRS Contributors). (2024). Free Spaced Repetition Scheduler Algorithm. *GitHub Repository*. https://github.com/open-spaced-repetition/free-spaced-repetition-scheduler

Zep / Graphiti Contributors. (2025). Zep: A Temporal Knowledge Graph Architecture for Agent Memory. *arXiv preprint arXiv:2501.13956*. Repository: https://github.com/getzep/graphiti

Works cited

1. JOHN T. WIXTED and EBBE B. EBBESEN - Brady Lab, UCSD, https://wixtedlab.ucsd.edu/publications/wixted/Wixted_and_Ebbesen_(1997).pdf
2. The form of the forgetting curve and the fate of memories, https://www.ampl-psych.com/wp-content/uploads/2021/05/Averell-and-Heathcote-2011-The-form-of-the-forgetting-curve-and-the-fate-of-m.pdf
3. A quantitative analysis of individual subject forgetting functions, https://www.researchgate.net/publication/13889717_Genuine_power_curves_in_forgetting_A_quantitative_analysis_of_individual_subject_forgetting_functions
4. Averaging Learning Curves Across and Within Participants, https://openresearch.newcastle.edu.au/ndownloader/files/54355223
5. One Hundred Years of Forgetting" A Quantitative Description of, https://users.cs.northwestern.edu/~paritosh/papers/KIP/100YearsOfForgetting.pdf
6. Forgetful but Faithful: A Cognitive Memory Architecture and, https://www.alphaxiv.org/abs/2512.12856
7. Ayush Gupta - AI Agent Engineer, https://www.madmecodes.in/blog/agentic-memory-landscape-2026
8. The Algorithm · open-spaced-repetition/awesome-fsrs Wiki - GitHub, https://github.com/open-spaced-repetition/awesome-fsrs/wiki/The-Algorithm
9. Free Spaced Repetition Scheduler - GitHub, https://github.com/open-spaced-repetition/free-spaced-repetition-scheduler
10. SqueakyRobot/fsrs: FSRS Implementation - GitHub, https://github.com/SqueakyRobot/fsrs
11. Benchmark of Spaced Repetition Algorithms, https://expertium.github.io/Benchmark.html
12. Long-Term Memory In Psychology: Types, Capacity & Duration, https://www.simplypsychology.org/long-term-memory.html
13. Fifty Years of Memory of College Grades: Accuracy and Distortions, https://fuse.franklin.edu/cgi/viewcontent.cgi?article=1110&context=facstaff-pub
14. Journal of Experimental Psychology: General - Ovid, https://www.ovid.com/journals/jepge/pdf/00004785-198403000-00001~semantic-memory-content-in-permastore-fifty-years-of-memory
15. fifty years of memory for Spanish learned in school - PubMed - NIH, https://pubmed.ncbi.nlm.nih.gov/6242406/
16. MemDream for Self-Probing Memory Evolution - arXiv, https://arxiv.org/pdf/2609.34545
17. Externalization in LLM Agents: A Unified Review of Memory, Skills, https://arxiv.org/html/2604.08224v1
18. MemDream for Self-Probing Memory Evolution - arXiv, https://arxiv.org/html/2609.34545v1
19. A Graph-Based Memory Framework for Long-Term LLM Agents - arXiv, https://arxiv.org/html/2608.28978v1
20. Forgetful but Faithful: A Cognitive Memory Architecture and ... - arXiv, https://arxiv.org/pdf/2512.12856
21. Control-Plane Placement Shapes Forgetting - arXiv,

https://arxiv.org/pdf/2606.15903
22. A Graph-Based Memory Framework for Long-Term LLM Agents - arXiv, https://arxiv.org/pdf/2608.28978
23. MEMCODEX: SELF-PROGRAMMING HIERARCHICAL MEMORY, https://openreview.net/pdf?id=tiVioB8acw
24. Memory OS of AI Agent - arXiv, https://arxiv.org/html/2506.06326v1
25. Memory of Memory - arXiv, https://arxiv.org/html/2609.25054v1
26. OpenAIGenericClient: json_schema response_format lets ... - GitHub, https://github.com/getzep/graphiti/issues/1909
27. Beyond Static Graphs: Engineering Evolving Relationships, https://www.getzep.com/blog/beyond-static-knowledge-graphs/
28. Graphiti Local by renezander030 | Glama, https://glama.ai/mcp/servers/renezander030/graphiti-local
29. [2512.12856] Forgetful but Faithful: A Cognitive Memory Architecture, https://arxiv.org/abs/2512.12856
30. GitHub - open-spaced-repetition/srs-benchmark, https://github.com/open-spaced-repetition/srs-benchmark
31. The Emergence of Psilocybin in Psychiatry and Neuroscience - MDPI, https://www.mdpi.com/1424-8247/18/4/555